

## WEEK - 3 PRACTICAL

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The screenshot shows the MongoDB Data Explorer interface. On the left, the 'CLUSTERS (1)' sidebar lists 'Cluster0' with sub-clusters 'admin', 'bookflow', 'book\_metadata', 'local', 'studentDB', and 'students'. The 'book\_metadata' cluster is selected. The main panel displays the 'Documents' tab for the 'book\_metadata' collection. A query is entered: `{ field: 'value' }`. The results show a single document with the following fields: `_id` (ObjectId), `book_id` (101), `title` ("The Martian"), `author` ("Andy Weir"), `published_year` (2021), `category` ("Science Fiction"), `format` ("Digital PDF"), `file_size` ("2.5 MB"), `member_id` (101), and `reviews` (Array (3)). The bottom status bar indicates 'Alert System Status: All Good' and '©2026 MongoDB, Inc.'.

The screenshot shows the MongoDB Data Explorer interface. On the left, the 'CLUSTERS (1)' sidebar lists 'Cluster0' with sub-clusters 'admin', 'bookflow', 'book\_metadata', 'local', 'studentDB', and 'students'. The 'book\_metadata' cluster is selected. The main panel displays the 'Documents' tab for the 'book\_metadata' collection. A query is entered: `{ field: 'value' }`. The results show a single document with the following fields: `_id` (ObjectId), `book_id` (105), `title` ("Data Structures"), `author` ("Mark Allen"), `published_year` (2024), `category` ("Computer Science"), `format` ("Digital PDF"), `file_size` ("5.0 MB"), `member_id` (105), and `reviews` (Array (2)). The bottom status bar indicates 'Alert System Status: All Good' and '©2026 MongoDB, Inc.'.

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

admin

bookflow

book\_metadata

local

studentDB

students

Cluster0 > bookflow > book\_metadata

View monitoring

Visualize Your Data

Documents 5

Aggregations

Schema

Indexes 1

Validation

{

"reviews.rating": {

"\$gt": 4

}

}

Explain

Reset

Find

Options

25

1 - 5 of 5

member\_id: 101

rating: 5

comments: "Excellent science fiction story"

1: Object (3)

member\_id: 102

rating: 4

comments: "Very interesting and informative"

2: Object (3)

member\_id: 103

rating: 5

comments: "Excellent space adventure"

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My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

admin

bookflow

book\_metadata

local

studentDB

students

Cluster0 > bookflow > book\_metadata

View monitoring

Visualize Your Data

Documents 5

Aggregations

Schema

Indexes 1

Validation

{

"reviews.rating": {

"\$in": [4, 5]

}

}

Explain

Reset

Find

Options

25

1 - 5 of 5

0: Object (3)

member\_id: 101

rating: 5

comments: "Excellent science fiction story"

1: Object (3)

member\_id: 102

rating: 4

comments: "Very interesting and informative"

2: Object (3)

member\_id: 103

rating: 5

comments: "Excellent space adventure"

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The screenshot displays the MongoDB Atlas Data Explorer interface. On the left sidebar, under "CLUSTERS (1)", the cluster "Cluster0" is expanded, showing databases "admin", "bookflow", and "studentDB". The "bookflow" database is selected, and the "book\_metadata" collection is highlighted.

The main panel shows the "Documents" tab for the "book\_metadata" collection. It contains a single document with the following JSON structure:

```
{
  "format": {
    "$regex": "Digital",
    "$options": "i"
  }
}
```

Below the document viewer, a table lists documents. The first document is visible:

| _id                                  | book_id | title         | author      | published_year | category          | format        | file_size | member_id |
|--------------------------------------|---------|---------------|-------------|----------------|-------------------|---------------|-----------|-----------|
| ObjectId('6a91432f6d442b6cd5cf3e80') | 101     | "The Martian" | "Andy Weir" | 2021           | "Science Fiction" | "Digital PDF" | "2.5 MB"  |           |

At the bottom of the interface, the "Alert System Status" is shown as "All Good". Footer links include ©2026 MongoDB, Inc., Status, Terms, Privacy, Atlas Blog, and Contact Sales.

Cluster0

book\_metadatadata

+

Cluster0 > bookflow > book\_metadatadata

Documents 5 Aggregations Schema Indexes 1 Validation

\$match

\$unwind

+

?

Explain

Run

Options

Untitled - modified 

+ CREATE NEW

 </>

PREVIEW 

{}

 </> ⚙

Stage 2 \$unwind 

☑

```
1 // **
2 * path: Path to the array field.
3 * includeArrayIndex: Optional name for
4 * preserveNullAndEmptyArrays: Optional
5 * toggle to unwind null and empty val
6 */
7 {
8   path: "$reviews"
9 }
10
```

Output preview after \$unwind stage (Sample of 10 documents)

```
_id: ObjectId('6a9143ae6d442b6cd5cf3e86')
book_id: 105
title: "Data Structures"
author: "Mark Allen"
published_year: 2024
category: "Computer Science"
format: "Digital PDF"
file_size: "5.0 MB"
member_id: 185
```

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Cluster0

book\_metadatadata

+

Cluster0 > bookflow > book\_metadatadata

Documents 5 Aggregations Schema Indexes 1 Validation

\$match

\$unwind

\$group

+

?

Explain

Run

Options

Untitled - modified 

+ CREATE NEW

 </>

PREVIEW 

{}

 </> ⚙

Stage 3 \$group 

☑

```
1 {
2   "_id": "$title",
3   "average_rating": {
4     "$avg": "$reviews.rating"
5   },
6   "total_reviews": {
7     "$sum": 1
8   }
9 }
```

Output preview after \$group stage (Sample of 4 documents)

```
_id: "Data Structures"
average_rating: 4
total_reviews: 2
```

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Cluster0

book\_metadatadata

+

Cluster0 > bookflow > book\_metadatadata

Documents 5 Aggregations Schema Indexes 1 Validation

\$match

\$unwind

\$group

\$sort

+

?

Explain

Run

Options

Untitled - modified 

+ CREATE NEW

 </>

PREVIEW 

{}

 </> ⚙

Stage 4 \$sort 

☑

```
1 |
2 {
3   "average_rating": -1
4 }
```

Output preview after \$sort stage (Sample of 4 documents)

```
_id: "Dune"
average_rating: 4.666666666666667
total_reviews: 3
```

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My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bookflow
  - book\_metadata
  - local
  - studentDB
    - students

Cluster0book\_metadata

Cluster0 > bookflow > book\_metadata

Documents5AggregationsSchemaIndexes2Validation

Create IndexRefresh

Looking for search indexes?  
These indexes can be created and viewed under [Search and Vector Search](#).

| Name & Definition       | Type    | Size    | Usage                     | Properties | Status |
|-------------------------|---------|---------|---------------------------|------------|--------|
| > _id_                  | REGULAR | 36.9 kB | 4 (since Fri Aug 28 2026) | UNIQUE     | READY  |
| > reviews.comments_text | TEXT    | 20.5 kB | 0 (since Fri Aug 28 2026) |            | READY  |

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My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bookflow
  - book\_metadata
  - local
  - studentDB
    - students

Cluster0book\_metadata

Cluster0 > bookflow > book\_metadata

Documents5AggregationsSchemaIndexes2Validation

{  
  "\$text": {  
    "\$search": "excellent"  
  }  
}

251 - 5 of 5

```
_id: ObjectId('6a91432f6d442b6cd5cf3e80')
book_id: 101
title: "The Martian"
author: "Andy Weir"
published_year: 2021
category: "Science Fiction"
format: "Digital PDF"
file_size: "2.5 MB"
member_id: 101
reviews: Array (3)
```

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## Explain Plan

Explain provides key execution metrics that help diagnose slow queries and optimize index usage. [Learn more](#)

Interpret

Visual Tree

Raw Output

```

{
  "signature": "TEXTSEARCH_QUERY",
  "parsedTextQuery": {
    "terms": ["excel"],
    "negatedTerms": [],
    "phrases": [],
    "negatedPhrases": []
  },
  "textIndexVersion": 3,
  "inputStage": {
    "stage": "FETCH",
    "inputStage": {
      "stage": "IXSCAN",
      "keyPattern": {
        "_fts": "text",
        "_ftsx": 1
      },
      "indexName": "reviews.comments_text",
      "isMultiKey": true,
      "isUnique": false,
      "isSparse": false,
      "isPartial": false,
      "indexVersion": 2,
      "isAscend": true,
      "isForward": true
    }
  }
}
```

### Query Performance Summary

5 documents returned

5 documents examined

0 ms execution time

Is not sorted in memory

5 index keys examined

Query used the following index:

\_fts (text)

\_ftsx ↑

Close